

Jungyu Kim

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EDUCATION

Mar.2021 – Present Kyonggi University Suwon, Korea
Bachelor of Mechanical Systems Engineering, Intelligent Robotics
GPA 3.4/4.5

RESEARCH INTEREST

Robot Manipulator Control URDF Modeling & Simulation Motion Planning Imitation Learning

RESEARCH EXPERIENCES

Jan.2026 - Present RV Lab Kyonggi University, Korea
- Undergraduate Research Intern
- Conducted research in Robot Manipulator Control and Simulation
Role: Implemented RTDE-based real-time control and end-effector control for UR10e and Doosan manipulator
: Converted CAD files to URDF format for robot simulation
: Visualized and controlled manipulator in Isaac Simulation environment

PUBLICATIONS

[1] "Perspective Transform-Based Data Augmentation to Mitigate Minority Class Imbalance for Road Marking Semantic Segmentation",
The Proceeding of the 2026 ICROS Conference (3rd Author) 

AWARDS

Nov. 2022 3rd Prize, SW Imaginary Company Project Contest, Council of SW-Centered Universities 
Jul. 2022 1st Prize, BARUN Problem Solving Project, Kyonggi University 

PROJECTS

- 1. Organ Extraction Robot Project** Apr.2026 - Present
- Collaborated with external company on robotic surgery automation
- Developed a full motion engine sequence implementation system for surgical robot automation
Role: Converted robot CAD files to URDF format and visualized in Isaac Simulation for robot control
: Implemented complete cutting sequence as part of the full motion engine pipeline
(External Project)
- 2. Diffusion Policy-Based Control of Doosan Robot Manipulator** May.2026 - Present
- Conducting research on imitation learning-based robot manipulator control using Diffusion Policy
Role: Configured Doosan robot environment and implemented robot simulation environment in Isaac Simulation
: Developing Diffusion Policy-based control pipeline for robot manipulation tasks
(Undergraduate Research Intern Project)
- 3. VR Teleoperation-Based Imitation Learning Data Collection**  Jan.2026 - Apr.2026
- Participated in a VR teleoperation project for robotic manipulation using Isaac Sim and ACT policy learning
Role: Operated WebXR-based VR teleoperation system to collect demonstration data for robot grasping tasks
: Contributed to ACT policy training by providing demonstration episodes via hand-tracking-based manipulation
(Undergraduate Research Intern Project)
- 4. 6-DOF Robotic Arm Control and AI Vision Implementation**  Jul.2025 - Dec.2025
- Developed a robotic arm control and AI vision system using Yahboom Dofbot SE (6-DOF)
Role: Assembled hardware and configured full development environment (VMware, Ubuntu, ROS)
: Implemented and executed robotic arm control code using Python and ROS
(Extracurricular Activities)
- 5. Autonomous Driving Project using Xycar Simulator**  Jul.2022 - Sep.2022
- Achieved track completion within 3 minutes using lane detection only
Role: Converted camera views to bird's eye view and generated binary images using the Sliding block algorithm
: Achieved stable driving on curved lanes through the Probabilistic Hough Transform
: Controlled vehicle speed and steering angle via ROS topics and Python scripting
(Extracurricular Activities)

6. Vehicle with Object Detection and Automatic Reporting System based on Deep Learning Mar.2022 - Jun.2022

- Developed an obstacle avoidance vehicle using Jetson Nano, ROS, and Python

Role: Created an automatic reporting system employing shock detection sensors and the Kakao API

: Implement dynamic objects detection using OpenCV and YOLO algorithms

(1st Prize, BARUN Problem Solving Project)

7. Drone Delivery Service for Underdeveloped Areas Mar.2022 - Dec.2022

- Utilized Gazebo Simulator and attach a camera to the drone for UDP video streaming

- Employed QGroundControl and PX4 Software In The Loop (SITL) for drone control

Role: Drone 3D modeling and visualization using Autodesk Fusion 360

(3rd Prize, SW Imaginary Company Project Contest)

ADDITIONAL INFORMATION

Tools

Fusion AutoCAD Inventor

Skills

Python ROS2 Isaac Simulation Rviz2 MoveIt RTDE

Etc

Word PowerPoint Excel Latex GitHub Windows Ubuntu

Activities

Robot Operating & Autonomous Driving(RO:AD) Club (Mar.2022 - Present)